
HyperRec - Method

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1 Methodology

1.1 LightGCN Embedding Generation

The initial step involves generating user and item embeddings using the LightGCN(1), trained on user-item interaction data, represented as a graph with users and items as nodes, and interactions as edges. The model learns embeddings that capture the collaborative relationships between users and items and employs a graph convolutional network structure. The embedding propagation is formulated as:

$$\mathbf{e}_u^{(k+1)} = \text{aggregate} \left\{ \mathbf{e}_i^{(k)} : i \in \mathcal{N}_u \right\} \quad (1) \quad \mathbf{e}_i^{(k+1)} = \text{aggregate} \left\{ \mathbf{e}_u^{(k)} : u \in \mathcal{N}_i \right\} \quad (2)$$

where $\mathbf{e}_u^{(k)}$ and $\mathbf{e}_i^{(k)}$ are the embeddings of user u and item i at the k -th layer, and $\mathcal{N}_u$ and $\mathcal{N}_i$ denote the set of neighbors of user u and item i , respectively. The final embeddings are obtained by averaging the embeddings from all layers:

$$\mathbf{e}_u = \frac{1}{K+1} \sum_{k=0}^K \mathbf{e}_u^{(k)} \quad (3)$$

$$\mathbf{e}_i = \frac{1}{K+1} \sum_{k=0}^K \mathbf{e}_i^{(k)} \quad (4)$$

1.2 Vision-Language Feature Extraction

The visual embedding extraction is performed using the LlavaForConditionalGeneration model along with its associated processor. An 8-bit quantized version of the model is deployed on a GPU to optimize efficiency. For each item, a prompt—constructed from the item’s title and description—is used to guide the model in generating a meaningful embedding. The image and textual prompt are processed together, and the resulting pooled hidden-state representation is taken as the item’s visual embedding.

To enhance recommendation performance, visual cues are integrated with collaborative signals from LightGCN. Specifically, each item’s visual embedding from the VLM is fused with its LightGCN embedding. Before fusion, the VLM embedding $\mathbf{v}_i$ is passed through a projection layer to align its dimensionality with that of the LightGCN embedding:

$$\mathbf{v}'_i = \text{Projector}(\mathbf{v}_i) \quad (5) \quad \mathbf{e}'_i = \alpha \mathbf{e}_i + (1 - \alpha) \mathbf{v}'_i \quad (6)$$

Here, α is a learnable parameter that controls the contribution of each embedding in the fusion process. The final user-item interaction score is computed as the dot product between the user embedding $\mathbf{e}_u$ and the fused item embedding $\mathbf{e}'_i$:

$$\hat{y}_{ui} = \mathbf{e}_u^\top \mathbf{e}'_i \quad (7)$$

The LightGCN model is trained independently, and the checkpoint with the lowest validation loss is saved. Separately, the LLaVA model generates item-level visual embeddings using a predefined prompt template. These embeddings are computed offline and stored, ensuring that the computationally intensive visual processing is only performed once. This design separates the heavy VLM

embedding generation from the frequent task of recommendation, making the overall system more efficient and sustainable.

1.3 Framework

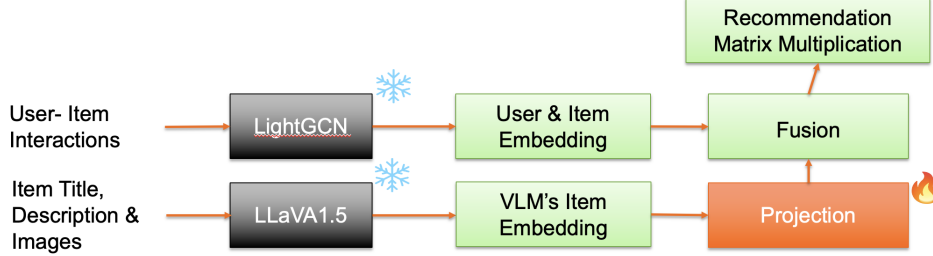


Figure 1: HyperRec Framework

1.4 Novelties and Contributions

We introduce **HyperRec**, a hybrid recommendation framework that enhances collaborative filtering with visual information extracted from vision-language models (VLMs). The key novelties and contributions of this work are as follows:

- **Multi-Modal Fusion:** HyperRec combines LightGCN-derived user-item embeddings with visual embeddings from VLMs, effectively leveraging both collaborative and semantic visual signals for more comprehensive item representations.
- **Adaptive Fusion Mechanism:** A learnable parameter α is employed to control the contribution of each modality during fusion, enabling the model to dynamically balance collaborative and visual information based on their relevance.
- **VLM Model Comparison:** HyperRec serves as a platform for systematically comparing different VLMs (e.g., LLaVA, SMOL) in the recommendation context. This comparative analysis provides insights into the representational strengths of various VLMs when applied to recommendation tasks.
- **Extensible Framework:** The architecture is modular and general, allowing easy integration of additional modalities beyond vision (e.g., textual, auditory), making HyperRec adaptable to diverse recommendation scenarios.

By addressing the limitations of traditional collaborative filtering—which often fails to capture item attractiveness or nuanced user preferences—HyperRec demonstrates how visual information can significantly enhance recommendation quality. Furthermore, our comparative study of VLMs highlights their varying effectiveness, contributing to a deeper understanding of how visual semantics influence recommendation performance.

2 Experiment Results

As part of my ongoing research, I plan to conduct experiments comparing the performance of HyperRec using different VLM for embeddings. I anticipate presenting the results in a tabular format, such as the example below:

Table 1: Performance Comparison of HyperRec with Different VLM Embeddings

VLM Model	Recall@10	NDCG@10
LLaVA-1.5-7B-HF	0.0175	0.007
SmolVLM2-2.2B-Instruct	TBD	TBD
CLIP (ViT-B/32)	TBD	TBD

References

- [1] Xiangnan He, Kuan Deng, Xiang Wang, Yan Li, Yongdong Zhang, and Meng Wang. LightGCN: Simplifying and powering graph convolution network for recommendation. In *Proceedings of the 43rd International ACM SIGIR Conference on Research and Development in Information Retrieval*, pages 639–648, 2020.
- [2] Steffen Rendle, Christoph Freudenthaler, Zeno Gantner, and Lars Schmidt-Thieme. BPR: Bayesian Personalized Ranking from Implicit Feedback. In *UAI*, pages 452–461, 2009.
- [3] Haotian Liu, Chunyuan Li, Qingyang Wu, and Yong Jae Lee. Improved Baselines with Visual Instruction Tuning. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2024.

A LLaVA Prompt Template

The following prompt template was utilized to generate descriptive text embeddings from item images using the LLaVA model[cite: 6]:

```
<image>Generate a short text description that captures the essence of  
Item titled [{item_title}] for use in a recommendation system. Item  
Description [ {item_desc} ]. Aim for a concise description that  
highlights key features, style, and use cases.  
?
```

Where {item_title} and {item_desc} are dynamically replaced with the title and description of the item from the metadata[cite: 7].